torch.quantile#

https://pytorch.org/docs/stable/generated/torch.quantile.html#torch.quantile

Description:

Computes the q-th quantiles of each row of the input tensor along the dimension dim. To compute the quantile, we map q in [0, 1] to the range of indices [0, n] to find the location of the quantile in the sorted input. If the quantile lies between two data points a < b with indices i and j in the sorted order, result is computed according to the given interpolation method as follows: linear: a + (b - a) * fraction, where fraction is the fractional part of the computed quantile index.

Function Signature

```
torch.quantile(input, q, dim=None, keepdim=False, *,
interpolation='linear', out=None) → Tensor#

Parameters

Keyword Arguments
```

Parameters

`Keyword`

interpolation (str, optional) – interpolation method to use when the desired quantile lies between two data points. Can be linear, lower, higher, midpoint and nearest. Default is linear. out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(2, 3)
>>> a
tensor([[ 0.0795, -1.2117,  0.9765],
        [ 1.1707,  0.6706,  0.4884]])
>>> q = torch.tensor([0.25, 0.5, 0.75])
>>> torch.quantile(a, q, dim=1, keepdim=True)
tensor([[[-0.5661],
        [ 0.5795]],

        [[ 0.0795],
        [ 0.6706]],

        [[ 0.5280],
        [ 0.9206]]])
>>> torch.quantile(a, q, dim=1, keepdim=True).shape
torch.Size([3, 2, 1])
>>> a = torch.arange(4.)
>>> a
tensor([0., 1., 2., 3.])
>>> torch.quantile(a, 0.6, interpolation='linear')
tensor(1.8000)
>>> torch.quantile(a, 0.6, interpolation='lower')
tensor(1.)
>>> torch.quantile(a, 0.6, interpolation='higher')
tensor(2.)
>>> torch.quantile(a, 0.6, interpolation='midpoint')
tensor(1.5000)
>>> torch.quantile(a, 0.6, interpolation='nearest')
tensor(2.)
>>> torch.quantile(a, 0.4, interpolation='nearest')
tensor(1.)
```

Notes & Warnings

> **NOTE** Note By default dim is None resulting in the input tensor being flattened before computation.

Detailed Documentation

Documentation

Computes the q-th quantiles of each row of the input tensor along the dimension dim.

To compute the quantile, we map q in [0, 1] to the range of indices [0, n] to find the location of the quantile in the sorted input. If the quantile lies between two data points a < b with indices i and j in the sorted order, result is computed according to the given interpolation method as follows:

linear: a + (b - a) * fraction, where fraction is the fractional part of the computed quantile index.

nearest: a or b, whichever's index is closer to the computed quantile index (rounding down for .5 fractions).

midpoint: (a + b) / 2.

If q is a 1D tensor, the first dimension of the output represents the quantiles and has size equal to the size of q, the remaining dimensions are what remains from the reduction.

By default dim is None resulting in the input tensor being flattened before computation.

input (Tensor) – the input tensor.

q (float or Tensor) – a scalar or 1D tensor of values in the range [0, 1].

dim (int, optional) – the dimension to reduce.

keepdim (bool, optional) – whether the output tensor has dim retained or not. Default: False.

interpolation (str, optional) – interpolation method to use when the desired quantile lies between two data points. Can be linear, lower, higher, midpoint and nearest. Default is linear.

out (Tensor, optional) – the output tensor.
